MILITARY SPECIFICATION SHEET

ELECTRON TUBE, PHOTOTUBE

TYPE CE-75V

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Red-infrared sensitive, with a spectral response of S1.

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ebb	Light	Ib	Rp	T	Light source
Unit:	Vdc	£m	μAdc	Meg	°C	---
Maximum:	250	---	15	---	100	---
TEST CONDITIONS:	250	0.1	---	1.0	---	LS-1 (see note 1)

GENERAL:

Qualification - Required

Ⓒ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
5301	Anode current	5		Ib	2.0	4.0	µAdc
5301	Dark current	5		Ib	---	0.00005	µAdc
---	Gas ratio	3, 5		Gr	---	1.5	---
	Quality conformance inspection, part 2						
1101	Secureness of base, cap, or insert	2		---	---	---	---
1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	4	Group C; Ebb = 250 Vdc; t = 500 hours (min)	---	---	---	---
---	Life-test end point:						
5301	Anode current	-		Ib	1.0	---	µAdc
(C) 1031	Low-frequency vibration (25 Hz)	6	No voltages applied	---	---	---	---
(C) 1331	Direct-interelectrode capacitance	6		Cpk	1.6	3.0	pF
(C) 5303	Spectral response identification	6	Si	---	---	---	---

NOTES:

1. The light during test, except in the case of life-test operating conditions, shall be incident upon a 0.5-inch diameter aperture whose center is located 2.13 inches above bottom edge of base, centered with respect to axis of tube.
2. Immersion in water shall be omitted and tubes shall be tested dry. Torque shall be applied with 20.0 lb-in. minimum for base, 1.5 lb-in. for cap.
3. Gas ratio = (Ib at Ebb = 250 Vdc)/(Ib at Ebb = 25 Vdc).
4. The luminous flux at the cathode shall be adjusted to give an average current of 9.0 microamperes on an average tube. Color temperature shall be greater than 2,000° K.
5. This test to be performed at the conclusion of the holding period.
6. (C) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

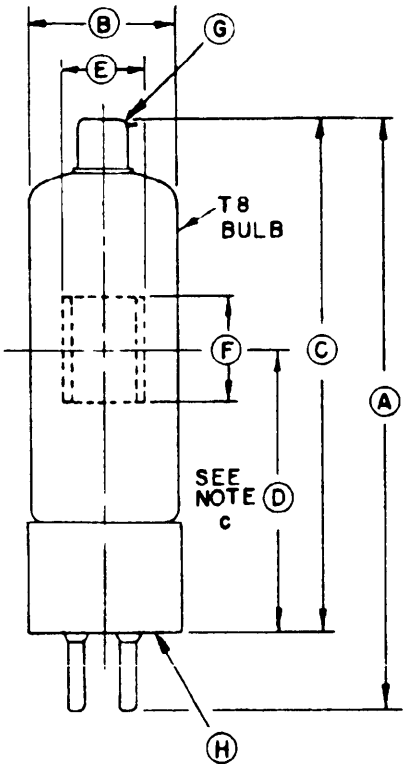
User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

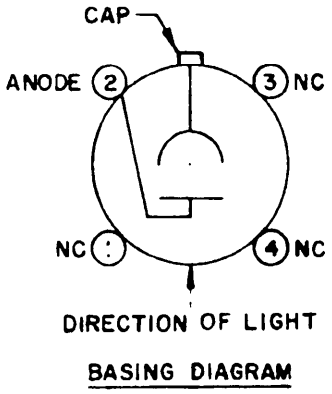
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3063)

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BASING DIAGRAM

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		4.500 (114.30)
Quality conformance inspection, part 3 (see note 6)		
B		1.190 (30.23)
C		3.875 (98.43)
D	2.031 (51.59)	2.219 (56.36)
E	.625 (15.88)	
F	.812 (20.62)	
G	Cap: C1-1	
H	Base: A4-5 or A4-26 (see note c)	

NOTES:
a. Dimensions D, E, and F apply to usable cathode area.
b. For pin alinement, use gage GA4-1.
c. Dimension D locates centerline of cathode.

FIGURE 1. Outline drawing of electron tube type CE-75V.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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